

Aditya Mukhopadhyay

AI Engineer – Badho Technologies
adul.m.2003@gmail.com | +91-9981861975 | Portfolio | LinkedIn | GitHub

EDUCATION

B.Tech, Information Technology
IGEC Sagar (RGPV), Madhya Pradesh

2021 – 2025 · CGPA 7.5

Class XII (PCM)

St. Joseph's Convent S.S. School,
Ranjhi, Jabalpur
2021 · 93.4%

LINKS

Portfolio

adityamukhopadhyay.vercel.app

LinkedIn

adityamukhopadhyay2003

GitHub

adityamukhopadhyay

SKILLS

Languages & Backend

Python (pandas, NumPy, scikit-learn) · SQL · FastAPI · REST APIs · GraphQL (Hasura) · PostgreSQL · Supabase · Docker · AWS (Lambda, EC2, S3) · Git

AI / ML

Machine Learning · LLM agents · Claude Agent SDK · Model Context Protocol (MCP) · multi-agent orchestration · tool calling · RAG · vector search (FAISS) · embeddings · semantic search · clustering (HDBSCAN, K-Means) · prompt engineering · evaluation & guardrails · cost tracing · NLP · deep learning (CNNs) · Anthropic Claude, OpenAI GPT and Google Gemini APIs · n8n

Leadership

Managed a 3-person AI team · worked directly with stakeholders · client-facing delivery

SUMMARY

AI Engineer who ships production large language model (LLM) agent systems end to end – multi-agent orchestration on the Claude Agent SDK, MCP tool servers, retrieval-augmented generation (RAG) and vector search, evaluation and guardrails, and classical ML – in Python, FastAPI and PostgreSQL on AWS.

WORK EXPERIENCE

AI Engineer | Badho Technologies Pvt. Ltd.

Jun 2025 – Present

- **Autonomous Delivery Fleet – multi-agent orchestration (Claude Agent SDK, MCP, Python, PostgreSQL)**
 - Delivery failures surfaced only when a human noticed a stuck rider.
 - Architected a **per-rider orchestrator agent (~20 concurrent)** that delegates to messaging, escalation, judgement and route-planning sub-agents over one in-process MCP server, with **deterministic guards below the model**.
 - Runs **unattended 13-hour shifts** in production, backed by **23 offline test suites** and an "Ask the fleet" agent for operators.
- **WhatsApp AI Sales Agent on Buyer MCP – conversational agent + the buyer app as 100 typed tools (Claude Agent SDK, MCP, Python, Hasura GraphQL, OAuth/JWT)**
 - Retailers asked questions and abandoned carts faster than a sales team could respond, and agents had no safe, permissioned way to act inside the buyer app.
 - Built an MCP server exposing search, carts, coupons, orders, wallet and tickets as **100 typed tools** behind phone-OTP OAuth and **per-buyer JWTs**, and on it a two-model agent: a tool-using worker plus a **tool-less reply agent with a claim verifier**, per-buyer queues and a bounded concurrency gate.
 - Handles buyer conversations end to end in production and influenced **475 orders (INR 5.0 lakh)**; the same MCP powers Claude Desktop and Claude Code, with the tool schema as the permission boundary.
- **Courier Ops Autopilot – claims & ticket automation (FastAPI, remote MCP, Claude Haiku, vision)**
 - Money went unclaimed because weight-overcharge and failed-delivery filing with the courier was manual.
 - Created a backend, dashboard and bearer-gated remote MCP over the courier API that **auto-files weight claims from sorter images**, raises NDR / lost / damaged tickets and drafts vendor replies, behind a **single liveness gate**.
 - Filed **698 claims and 370 tickets in the first 24 hours; 2,143 closed** to date.
- **Search, Matching & Catalog Intelligence – ML pipeline (FAISS, HDBSCAN, embeddings, PostgreSQL, n8n)**
 - Hinglish and phonetic searches missed, **85,000+ SKUs** had no variant grouping, and **170,000+ competitor SKUs** needed catalogue mapping.
 - **Led a 3-person team** to ship **sub-100 ms hybrid search** (FAISS + local embeddings + metaphone), sister-product clustering (MiniLM, HDBSCAN / K-Means, shadow-write staging with human approval) and a cost-tiered matching funnel (exact → trigram → batched LLM).
 - Mapped **~40,000 product groups**, lifted sister-product conversion from **20% to 60%**, and matched **170,000+ SKUs**.

AI Developer (Freelance) | Strix Digital

Jul 2023 – May 2025

- Delivered AI web apps (a **cold-email generator on Groq**), scrapers, automation tooling and an **iOS app** for a Houston client, client-facing from scoping to hand-off.

Data Science Intern | Acmegrade (in collab. with IIT Bombay)

Apr 2022 – Jun 2022

- Built ML models for **movie recommendation**, **sales forecasting** and **CNN sound classification**, with EDA and feature engineering on multi-modal datasets.